

## TASK 11: File Handling – User Data Storage System

### Tools:

- Primary: IntelliJ IDEA, Java IO & NIO libraries
- Alternatives: Eclipse, VS Code

### Hints / Mini Guide:

1. Create a Java console application that allows users to enter personal details such as name, email, and phone number.
2. Use File, FileWriter, and BufferedWriter classes to store user data permanently in a text file.
3. Implement file reading using FileReader and BufferedReader to retrieve stored data.
4. Structure data in a readable format so that records can be easily interpreted.
5. Handle scenarios where the file does not exist by creating it dynamically.
6. Add exception handling to manage IO failures and permission issues.
7. Demonstrate appending data instead of overwriting existing data.
8. Close all file resources properly to avoid memory leaks.

### Deliverables:

- Text file containing stored user records
- Java program demonstrating read/write operations

### Interview Questions Related To Above Task:

- Difference between FileReader and BufferedReader?
- Why buffering improves performance?
- What is IOException?
- Difference between IO and NIO?
- How to avoid resource leaks?

## 📌 Task Submission Guidelines

- 🕒 **Time Window:**

You can complete the task anytime between 10:00 AM to 10:00 PM on the given day. Submission link closes at 10:00 PM

- 🔍 **Self-Research Allowed:**

You are free to explore, Google, or refer to tutorials to understand concepts and complete the task effectively.

- 🔧 **Debug Yourself:**

Try to resolve all errors by yourself. This helps you learn problem-solving and ensures you don't face the same issues in future tasks.

- 💰 **No Paid Tools:**

If the task involves any paid software/tools, do not purchase anything. Just learn the process or find free alternatives.

- 📁 **GitHub Submission:**

Create a new GitHub repository for each task.

Add everything you used for the task — code, datasets, screenshots (if any), and a short README.md explaining what you did.

- 📌 **Submit Here:**

After completing the task, paste your GitHub repo link and submit it using the link below:

- 👉 [[Submission Link](#)]

Best  
of  
Luck

